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General Quantification of Covariate and Concept Shifts

Entropic Optimal Transport Yields the First General, Estimable Error Bound Unifying Both Shift Types

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Generalization under distribution shift is a core unsolved challenge in modern machine learning. Existing learning bound theory handles covariate shift and concept shift in separate, narrow, idealized frameworks — and both frameworks break when the source and target distributions do not share the same support, which is the common case in deployment. This paper derives a single general bound that unifies both shift types using entropic optimal transport, is estimable from samples, and comes with an algorithm — DataShifts — that measures each component of the shift from finite data.

AT A GLANCE

Problem: Existing distribution-shift bounds are restricted to matching supports and separate shift types; neither condition holds in practice.

Why it matters: Unestimable, vacuous, or type-specific bounds cannot guide real deployment decisions under shift.

Method: Define γ^* -concept shifts via entropic OT; derive a unified bound; construct sample estimators for both components.

Result: DataShifts produces finite non-vacuous bounds on all 12 synthetic and 5 real benchmarks; 1.5–5 $\times$ tighter than prior methods where prior methods are finite.

Evidence: Formal concentration guarantees; ablations on SCM size, sample size, and cost function.

The Core Problem

A learning algorithm is trained on source data $(X_s, Y_s) \sim P_s$ and deployed on target data $(X_t, Y_t) \sim P_t$. How much does the test error on P_t exceed the training error on P_s ?

Covariate shift assumes the label mechanism is unchanged ($P_t(Y|X) = P_s(Y|X)$) but the input distribution shifts ($P_t(X) \neq P_s(X)$). Importance-weighted ERM corrects this.

Concept shift assumes the conditional itself changes: $P_t(Y|X) \neq P_s(Y|X)$. Existing bounds measure this change via f -divergences between conditionals — but these divergences are infinite when $\text{supp}(P_t) \not\subseteq \text{supp}(P_s)$, producing vacuous bounds in the mismatch setting that characterizes most real deployments.

Why Existing Bounds Fail

The authors prove a formal impossibility: any bound of the form

$$\mathcal{E}_t(h) \leq \mathcal{E}_s(h) + D_f(P_s, P_t) + \lambda$$

where D_f is any f -divergence-based concept shift measure will be infinite whenever $\text{supp}(P_t) \not\subseteq \text{supp}(P_s)$, even if the conditional distributions are in practice very close.

This is not merely a theoretical inconvenience. On standard domain adaptation benchmarks such as Camelyon17 (where hospital scanners differ) or CIFAR-10-C (where corruptions shift the input distribution globally), the target support extends beyond the source support. All prior bounds are vacuous on these benchmarks.

Entropic Optimal Transport and γ^* -Concept Shifts

The key insight is to replace f -divergence with a soft coupling via entropic optimal transport. Given a cost function $c(x, x')$ on the input space, define:

$$\gamma^* = \arg \min_{\gamma \in \Pi(P_s^X, P_t^X)} \mathbb{E}_\gamma[c(X, X')] + \varepsilon H(\gamma)$$

where $H(\gamma)$ is the entropy of the coupling and $\varepsilon > 0$ is a regularization parameter. The γ^* -concept shift is:

$$\Delta_\gamma = \mathbb{E}_{\gamma^*}[\text{TV}(P_s(Y|X=x), P_t(Y|X'=x'))]$$

This measures how much the label distribution changes between “matched” source and target points under the optimal transport plan. When supports agree, this recovers the classical TV-based concept shift. When supports mismatch, the transport plan couples each target point to the nearest available source point, producing a well-defined, finite quantity.

The General Error Bound

Theorem. For any hypothesis h with bounded loss $\ell \in [0, 1]$:

$$\mathcal{E}_t(h) \leq \underbrace{\mathcal{E}_s(h) + W_\varepsilon(P_s^X, P_t^X)}_{\text{covariate shift}} + \underbrace{\Delta_\gamma(P_s, P_t)}_{\gamma^*\text{-concept shift}} + O\left(\sqrt{\frac{\log |\mathcal{H}|}{n}}\right)$$

where W_ε is the entropic Wasserstein distance between input marginals. This bound:

- Applies to multiclass, regression, and stochastic labeling.
- Is finite for arbitrary source/target supports.
- Recovers known tight bounds as special cases.
- Is estimable from finite samples.

The DataShifts Algorithm

DataShifts estimates both shift components from source and target samples:

Step 1. Solve $\hat{\gamma}^* = \text{Sinkhorn}(\hat{P}_s^X, \hat{P}_t^X, c, \varepsilon)$ via Sinkhorn iterations.

Step 2. Compute $\hat{W}_\varepsilon$ as the primal OT objective.

Step 3. For each matched pair (x_i, x'_j) in $\hat{\gamma}^*$, estimate $\text{TV}(P_s(Y|x_i), P_t(Y|x'_j))$ from label empiricals.

Concentration guarantee: With probability $1 - \delta$,

$$|\hat{W}_\varepsilon - W_\varepsilon| \leq C \sqrt{\frac{\log(1/\delta)}{n}}$$

and similarly for $\hat{\Delta}_\gamma$, where C depends on the diameter of the input space.

Experimental Findings

Experiments span 12 synthetic shift scenarios and 5 real-world benchmarks (Portraits, CIFAR-10-C, Camelyon17, DomainNet, and a tabular clinical dataset):

Dataset	$\mathcal{H}\Delta\mathcal{H}$ bound	DataShifts
CIFAR-10-C (blur)	∞	0.48 (true: 0.31)
Camelyon17	2.31	0.39 (true: 0.24)
Portraits	0.87	0.27 (true: 0.18)

DataShifts is 1.5–5× tighter where prior methods are finite, and produces finite bounds where prior methods give ∞ .

Shift Decomposition

A key practical capability of DataShifts is decomposing total shift into covariate and concept components, guiding which correction strategy to apply:

CIFAR-10-C (corruption-only): $\hat{W}_\varepsilon = 0.41$, $\hat{\Delta}_\gamma = 0.03$ — shift is almost entirely covariate; apply importance weighting.

Clinical tabular (changed deployment policy): $\hat{W}_\varepsilon = 0.07$, $\hat{\Delta}_\gamma = 0.28$ — shift is almost entirely concept; apply label calibration.

Ablations and Interpretation

Regularization ε : Smaller values tighten bounds but increase optimization difficulty; $\varepsilon = 0.1$ works well across all tested datasets.

Sample size: DataShifts requires roughly $n = 500$ target samples for reliable bounds; $n = 100$ suffices for decomposition quality.

Cost function: Euclidean cost works for image benchmarks; learned feature-space cost improves results on tabular data.

Reference

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